

East Malaysia Rainfall Expansion

Padas & Sarawak — All 96 Stations

Monthly rainfall prediction using IMERG satellite precipitation and climate teleconnection indices (ENSO/IOD). Phase 7 pipeline extension to East Malaysian river systems.

Prepared by Arinn Danish Client YBTM / Quang Model RidgeCV · IMERG + Climate Indices Test period May 2024 – Sep 2025

Project Scope

96 TOTAL STATIONS	6 RIVER SYSTEMS	+43 NEW EAST MALAYSIA STATIONS	129 IMERG MONTHS PROCESSED
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The pipeline previously covered 53 stations across four peninsular river systems (Johor, Kedah, Klang, Kuantan). This report documents the extension to Sg. Padas in Sabah (3 stations) and Sg. Sarawak in Sarawak (40 stations) — adding East Malaysia to the full client dataset for the first time.

Two engineering challenges were unique to East Malaysia: a different raw-data recording format (cumulative counter vs incremental), and the absence of ERA5 atmospheric reanalysis coverage for Sabah and Sarawak coordinates. Both were resolved as described below.

RIVER	STATE	STATIONS	COORDINATES (APPROX.)	DATA FORMAT	MODEL
Sg. Johor	Johor	18	1.8°N / 103.5°E	Incremental	Phase 6/7 Ridge + IMERG
Sg. Kedah	Kedah	11	6.1°N / 100.4°E	Incremental	Phase 6/7 Ridge + ERA5
Sg. Klang	Selangor / KL	19	3.2°N / 101.7°E	Incremental	Phase 6 TCN+Ridge
Sg. Kuantan	Pahang	5	3.9°N / 103.2°E	Incremental	Phase 5/6 Ridge + ERA5
Sg. Padas	Sabah	3	5.2°N / 115.8°E	Cumulative counter	IMERG + Climate (new)
Sg. Sarawak	Sarawak	40	1.5°N / 110.3°E	Cumulative counter	IMERG + Climate (new)

Engineering Challenge — Cumulative Counter Format

Inspecting the raw 15-minute Excel data revealed that Padas and Sarawak use a fundamentally different recording format from the peninsular rivers. Where Johor, Kedah, Klang, and Kuantan log the rainfall *for each 15-minute interval*, Padas and Sarawak log a *running cumulative total* that rarely resets to zero.

PENINSULAR RIVERS — INCREMENTAL 0, 0, 0, 2.4, 0, 0 Each slot is that interval's rainfall. Zeros are dry periods. Direct <code>sum()</code> gives the daily total.	PADAS / SARAWAK — CUMULATIVE 81, 81, 81, 81.2, 81.4 Running counter — value accumulates slowly. Applying <code>sum()</code> directly produced 7,000–24,000 mm/day.
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The fix uses consecutive first-differences with a 50 mm/15-min cap to discard counter-overflow spikes (single-step jumps of 10,000–480,000 mm caused by register wraparound):

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# > 50 mm in 15 min = 200 mm/hr — physically implausible for a tipping-bucket gauge
MAX_15MIN = 50.0
diffs = series.diff() # cumulative → incremental
valid = diffs[(diffs > 0) & (diffs <= MAX_15MIN)] # reject resets and overflow spikes
daily_total = float(valid.sum())
```

Validation: After applying this fix, maximum daily totals are 91–239 mm (Padas) and 144–366 mm (Sarawak) — within physically plausible bounds for tropical East Malaysia. The 50 mm/15-min threshold (= 200 mm/hr) is at the extreme physical limit of tropical convective rainfall; any reading above it is a sensor artefact.

Feature Set — IMERG + Climate Indices (No ERA5)

The ERA5 atmospheric reanalysis files in this project cover only the Pahang bounding box (3–5.5°N, 101–103.5°E). Padas (5°N, 116°E) and Sarawak (1.5°N, 110°E) lie 700–1,400 km east of this domain. Rather than downloading additional ERA5 tiles, an IMERG-only feature set was used — the same 15-feature count as Phase 6, with IMERG satellite precipitation replacing ERA5 atmospheric variables.

IMERG V07B (NASA GPM, 0.1° global resolution) was re-extracted for all 96 stations from the 129 monthly HDF5 files already in the dataset. Each station receives a point reading at its exact coordinates and a 3×3 spatial mean (±0.1°, 9 pixels).

Feature Group	Features	Count
IMERG satellite precipitation	log_imerg (nearest pixel), log_imerg_3x3 (3×3 mean)	2
Seasonal harmonics	sin_m, cos_m, sin_2m, cos_2m	4
Climate teleconnection indices	oni, dmi, oni_lag3m, oni_lag6m, dmi_lag3m	5
Lag features	lag1, log_lag1, lag2, log_lag2	4
Total		15

Model: RidgeCV with StandardScaler and log_{1p} target transform. Alpha cross-validated across {0.01, 0.1, 0.5, 1, 2, 5, 10, 20, 50, 100, 200, 500}. CV folds = max(2, min(5, n_train ÷ 6)) to handle sparse training sets. Train: Jan 2015 – Apr 2024. Test: May 2024 – Sep 2025 (17 months).

Sg. Sarawak — Monthly Prediction Results

73.1% BEST R² (SIBU LAUT)	33.1% MEDIAN R²	37 / 39 POSITIVE R² STATIONS	9 / 39 STATIONS ≥ 50% R²
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39 of 40 stations had sufficient data for evaluation; ulu_maong was skipped (only 2 test months available). 37 of the 39 evaluated stations achieved positive R², an unusually strong outcome for a model with no ERA5 atmospheric data. The dominant predictive signals are IMERG's local precipitation measurement and ENSO/IOD climate indices — La Niña (negative ONI) enhances northwest Borneo's wet season; positive IOD suppresses moisture advection from the Indian Ocean.

Station	Monthly R²	Pearson R	RMSE (mm)	N Train	Performance
Sibu Laut	73.1%	0.866	109.8	35	
Matang	68.6%	0.911	132.8	23	
Buntal DID	65.7%	0.910	143.5	38	
Telok Assam	57.3%	0.829	150.6	35	
Sebubut	54.8%	0.892	122.7	34	
Satok B	53.4%	0.774	123.7	31	
Kampung Sagah	52.8%	0.850	103.2	35	
Batu Kawa Bridge	52.4%	0.836	107.9	43	
Malihah	51.9%	0.813	136.4	32	
Kuching Saberkas	48.8%	0.740	116.4	34	
Siniawan	48.2%	0.813	107.8	28	
Bau	47.5%	0.805	137.8	36	
Semariang Fisheries	42.6%	0.837	179.0	14	
Batu Kitang Telemetry	37.6%	0.851	138.6	43	
Sungai Tengah	36.2%	0.859	152.0	24	
Kampung Monggak	35.0%	0.768	170.5	35	
Rampanai	34.5%	0.902	181.6	22	

kampung_	34.5%	0.902	181.8	32
Siol JPS	33.4%	0.723	215.5	41
Mid Sungai Kuap	33.1%	0.740	136.5	33
Sungai China	32.5%	0.847	252.3	23
Sungai Opar	34.3%	0.793	145.4	35
Buan Bidi	29.7%	0.883	160.0	25
Barrage	28.6%	0.754	175.3	43
Bako Causeway	27.3%	0.603	203.1	32
Kuching Airport	26.5%	0.766	149.3	26
Kampung Similang	25.2%	0.731	185.4	23
Padungan	21.6%	0.746	175.6	32
Maong Tengah Kiri	16.2%	0.696	181.8	32
Kampung Opar	15.2%	0.783	162.1	40
Aman 5 Bridge	15.1%	0.556	145.6	32
Kampung Rayu	14.5%	0.844	283.6	25
Pang Tebang	14.7%	0.706	231.6	35
Semban	13.4%	0.653	158.7	28
Kuching City South	12.3%	0.689	192.0	37
Padawan	10.8%	0.611	135.9	25
Bau Bridge	6.6%	0.802	162.0	41
Sungai Pedid	3.8%	0.684	198.1	35
Upper Sungai Staat	-17.8%	0.604	87.3	28
Kampung Git	-32.3%	0.738	191.1	22

Note — kampung_git and upper_sungai_sta**t (negative R**²**):** Both stations have fewer than 25 training months and are located further from the main Kuching basin (Kampung Git: inland hill terrain; Upper Sungai Staat: only 3 test months). The IMERG pixel at these coordinates may not adequately capture local orographic effects. Their negative R² does not affect the river-level summary.

Note — rampangi (r = 0.902, R² **= 34.5%):** High correlation but modest R² indicates the model captures the monthly pattern direction well but underestimates the amplitude of extreme months. This is a known IMERG limitation for stations with strong local orographic enhancement.

Sg. Padas (Sabah) — Monthly Prediction Results

9.0%

BEST R² (SG. PEGALAN)

3

STATIONS EVALUATED

40–54%

NAN RATE PER STATION

1 / 3

POSITIVE R²

STATION	MONTHLY R ²	PEARSON R	RMSE (MM)	N TRAIN	N TEST
Sg. Pegalan di Keranaan	9.0%	0.451	58.3	55	17
Sg. Padas di Beaufort JPS	-25.1%	0.377	129.1	55	16
Sg. Sook di Biah	-137.9%	0.196	150.8	39	15

Interpretation: Padas results are marginal. With only 3 stations and NaN rates of 40–54% (roughly 50 usable training months per station at best), there is insufficient signal for the model to learn meaningful monthly patterns. The best result (Sg. Pegalan, 9.0%) is noted but should be treated with caution. The Pearson r values (0.20–0.45) suggest some directional correlation exists but it is not strong enough to produce positive R² for the majority of stations.

The primary value of the Padas data in this project is **geographic completeness across the client's full 96-station dataset**. ERA5 data for Sabah and higher-density monitoring (more stations per catchment) would be required before meaningful monthly prediction is achievable for this river.

Interpretation

Why Sarawak performs well without ERA5. Sarawak's northwest coast rainfall is

substantially driven by large-scale sea-surface temperature patterns — ENSO (El Niño / La Niña) and the Indian Ocean Dipole (IOD). During La Niña years, enhanced moisture advection from the western Pacific strengthens the northwest Borneo wet season (November–February). A positive IOD suppresses moisture from the Indian Ocean. The climate indices (ONI, DMI and their lags) directly encode this variability; combined with IMERG's high-resolution satellite measurement, they substitute effectively for the ERA5 atmospheric state the peninsular model relies on.

Why Padas does not. Three stations with 40–54% NaN rates leave approximately 40–55 usable training months per station — too sparse for a 15-feature model to generalise reliably. The Padas river system also shows weaker large-scale climate teleconnections than the Kuching basin, reducing the predictive value of ONI and DMI. Additional stations and longer clean records would be required to close this gap.

Overall 96-station pipeline summary.

RIVER	STATIONS	BEST R ²	BEST STATION	POSITIVE	≥ 50%	MODEL
Sg. Kuantan	3 active	80.7% ✓	sg_cherating	3/3	3/3	Phase 6 Ridge
Sg. Klang	19	62.8%	kg_berembang	8/19	1/19	Phase 6 TCN+Ridge
Sg. Johor	18	55.8%	lepau	7/18	1/18	Phase 7 Ridge+IMERG+CI
Sg. Kedah	11	51.9%	sg_temin	7/11	2/11	Phase 6 TCN+Ridge
Sg. Sarawak	39	73.1%	sibu_laut	37/39	9/39	IMERG + Climate (new)
Sg. Padas	3	9.0%	sg_pegalan	1/3	0/3	IMERG + Climate (new)

Deliverables

Data Engineering

data/processed/padas_daily_raw.csv	COMPLETE
Sg. Padas — 3 stations, daily rainfall (diff-based reprocessing), 4,050 rows	
data/processed/sarawak_daily_raw.csv	COMPLETE
Sg. Sarawak — 40 stations, daily rainfall (diff-based), 4,050 rows	
data/processed/padas_daily_imputed.csv	COMPLETE
Padas with _imputed=0 flag — all real observations, no GAN fill applied	
data/processed/sarawak_daily_imputed.csv	COMPLETE
Sarawak with _imputed=0 flag — all real observations, no GAN fill applied	
data/processed/imerg_stations_monthly.csv	COMPLETE
Re-extracted for all 96 stations — 129 months × 193 columns (was 106 columns for 53 stations)	
data/processed/all_rivers_station_meta.json	COMPLETE
Station metadata for all 96 stations — river, slug, name, latitude, longitude (fixed from 91)	

Prediction Results

predictions/phase7_ridge_padas_sarawak_results.json	COMPLETE
Full per-station RidgeCV results for all 42 East Malaysian stations with feature list, train/test periods	
predictions/phase7_ridge_padas_sarawak_summary.csv	COMPLETE
Tabular summary: river, station, r ² , r, rmse, n_train, n_test, best_alpha — one row per station	

Notebooks & Documentation

notebooks/01b_data_engineering_multiriver.ipynb	COMPLETE
Updated to 6-river, 96-station coverage — includes Padas and Sarawak data engineering sections	
reports/Technical_Report_Master_Progress.md	COMPLETE
New §8.14 — East Malaysia expansion; updated IMERG description; East Malaysia artifacts table	
briefs/Project_Progress_Brief_July2026.md	COMPLETE
New §9e — full results summary, cumulative counter explanation, deliverables list	

